

ARNA GHOSH

Computer Science, McGill University & MILA, Montréal, Canada

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Fifth year PhD student investigating intelligent systems by combining Artificial Intelligence and Neuroscience.

EDUCATION

McGill University & MILA, Montréal
Ph.D. Computer Science (Vanier Scholar)

September 2019 - Present
CGPA: 4.0/4.0

McGill University, Montréal
M.Sc. Neuroscience

September 2017 - August 2019
CGPA: 4.0/4.0

Indian Institute of Technology (IIT) Kharagpur

B.Tech. Electrical Engineering, Minor in Computer Science (Institute **Silver medal**)

July 2013 - May 2017
CGPA: 9.26/10.0

RELEVANT INDUSTRY EXPERIENCE

Research Scientist Intern, FAIR, Meta Reality Labs

May 2023 - Jul 2023

- Developed unsupervised metrics of representation quality to evaluate and improve *self-supervised learning* models for time series data, specifically electromyography (EMG) data.

AI Research Scientist Intern, Ctrl-labs, Meta Reality Labs

Aug 2022 - Dec 2022

- Surgical inspection of pretrained networks to investigate learned feature space and *few-shot learning* for rapidly adapting a suitable feature space for identifying new categories.

Scientist in Residence, Innodem Neurosciences

June 2020 - Feb 2021

- Developed a **Tensorflow** module implementing the *Deepdream* algorithm to visualize features extracted by eye gaze-prediction neural networks and to improve interpretability for application in healthcare industry.

Developer, CBrain, McGill Centre for Integrative Neuroscience

July 2019 - August 2019

- Designed a well-documented **Python** wrapper library around the CBrain web platform to run neuroimaging tasks on a high-performance cluster from a UNIX terminal.

PUBLICATIONS

Full Papers:

- R. Pogodin*, J. Cornford*, **A. Ghosh**, G. Gidel, G. Lajoie, B.A. Richards, *Synaptic Weight Distributions Depend on the Geometry of Plasticity*, 12th International Conference on Learning Representations (ICLR), Vienna, May 2024 [**spotlight**].
- P. Li*, J. Cornford*, **A. Ghosh**, B.A. Richards, *Learning better with Dale's law: a spectral perspective*, 37th Conference on Neural Information Processing Systems (NeurIPS), New Orleans, Dec 2023.
- Z. Chorghay, V.J. Li, A. Schohl, **A. Ghosh**, E.S. Ruthazer. *The effects of the NMDAR co-agonist D-serine on the structure and function of the optic tectum*. Scientific Reports, Aug 17 2023;13(1):13383.
- A. Ghosh**, Y.H. Liu, G. Lajoie, K.P. Körding, B.A. Richards, *How gradient estimator variance and bias impact learning in neural networks*, 11th International Conference on Learning Representations (ICLR), Kigali, Rwanda, May 2023.
- K.K. Agrawal*, A.K. Mondal*, **A. Ghosh***, B.A. Richards, *α -ReQ: Assessing representation quality by measuring eigenspectrum decay*, 36th Conference on Neural Information Processing Systems (NeurIPS), New Orleans, Dec 2022.

*Equal contribution

6. Y.H. Liu, **A. Ghosh**, B.A. Richards, E. Shea-Brown, G. Lajoie, *Beyond accuracy: generalization properties of biologically-plausible temporal credit assignment rules*, 36th Conference on Neural Information Processing Systems (NeurIPS), New Orleans, Dec 2022.
7. A. Xifra-Porxas*, **A. Ghosh***, G.D. Mitsis, M.H. Boudrias, *Estimating brain age from structural MRI and MEG data: Insights from dimensionality reduction techniques*, NeuroImage 231, 117822, May 2021.
8. **A. Ghosh**, F. Dal Maso, M. Roig, G.D. Mitsis, M.H. Boudrias, *Unfolding the effects of acute cardiovascular exercise on neural correlates of motor learning using Convolutional Neural Networks*, Frontiers in Neuroscience 13 (2019): 1215, November 2019.
9. **A. Ghosh**, S. Singh, D. Sheet, *Simultaneous Localization and Classification of Acute Lymphoblastic Leukemic Cells in Peripheral Blood Smears Using a Deep Convolutional Network with Average Pooling Layer*, IEEE 12th International Conference on Industrial & Information Systems (ICIIS), December 2017.

Key Poster Presentations:

1. **A. Ghosh**, K.K. Agrawal, Z. Chorghay, A.K. Mondal, B.A. Richards, *Neural Population Geometry across model scale: A tool for cross-species functional comparison of visual brain regions*, 20th Computational and Systems Neuroscience (COSYNE), Montreal, March 2023.
2. J. Cornford, **A. Ghosh**, G. Gidel, B.A. Richards, *Learning in neural networks with brain-inspired geometry*, 20th Computational and Systems Neuroscience (COSYNE), Montreal, March 2023.
3. D. Lin, R. da Silva, **A. Ghosh**, R. Tong, S. Trenholm, B.A. Richards. *Neuronal optimal stimuli synthesized with deep learning reveal functional segregations in the mouse visual cortex*. The Society for Neuroscience, 52nd Annual Meeting, San Diego, CA, November 2022.
4. **A. Ghosh**, K.P. Körding, B.A. Richards. *Approximate gradient descent and the brain: the role of bias and variance*. 19th Computational and Systems Neuroscience (Cosyne), Lisbon, March 2022.
5. **A. Ghosh***, K.K. Agrawal*, B.A. Richards. *Characterizing High Dimensional Representation Learning in Overparameterized Neural Networks*. Workshop on the Theory of Overparameterized Machine Learning, April 2021.
6. M. Samiei*, **A. Ghosh***, B.A. Richards. *Mimicking mammalian navigation in watermaze using brain-inspired representations*. 34th Annual Conference on Neural Information Processing Systems (NeurIPS) workshop: Biological and Artificial Reinforcement Learning, December 2020.
7. **A. Ghosh***, A. Xifra-Porxas*, G.D. Mitsis, M.H. Boudrias. *Combining structural MRI images and MEG recordings for Biological brain age prediction*, Organization of Human Brain Mapping (OHBM), Annual Meeting, June 2019. [Link](#)
8. **A. Ghosh***, A. Xifra-Porxas*, G.D. Mitsis, M.H. Boudrias. *Biological brain age prediction using structural MRI: Insights from dimensionality reduction techniques*, International Society for Magnetic Resonance in Medicine (ISMRM), Annual Meeting, May 2019. [Link](#)

MAJOR PROJECTS

Representation geometry in the brain and machines

Jan 2021 - Present

Doctoral thesis project, Supervisor: Dr. Blake A. Richards

- Characterizing the geometry of learned representations in artificial neural networks and connecting these properties to generalization performance.
- Using self-supervised learning neural network models of vision to develop hypotheses around the representation geometry in brains across different species.

Biologically-plausible credit assignment for representation learning

Sep 2019 - Present

Doctoral thesis project, Supervisor: Dr. Blake A. Richards

- Understanding the impact of imperfect credit assignment on performance of learning systems.
- Developing a biologically-plausible temporal credit assignment and a self-supervised learning algorithm for representation learning from continuous stream of visual experience.

Deep Learning for Neuroimaging data analysis

Sep 2017 - Aug 2019

Masters thesis project, Supervisors: Dr. Marie-Hélène Boudrias & Dr. Georgios D. Mitsis

- Developed a deep learning-based prediction framework and a network interpretability module in **Torch** to identify exercise-induced task signatures for electroencephalography (EEG) data.
- Designed a ML-based **brain age prediction** system from magnetic resonance imaging (MRI) and magnetoencephalography (MEG) recordings in **PyTorch**.

Brexting: Brain Texting

Dec 2017 - Aug 2018

Intel Innovate FPGA Grand Finalists (**Silver** & **Iron** awards in regional & grand finals resp.)

- Built a deep learning-powered brain-computer interface on an Intel FPGA board to enable real-time typing from imagined motor movements. Code available on github repo.

Deep Learning for mitotic figure detection

Jul 2016 - May 2017

Bachelor's Thesis Project, Supervisor: Dr. Debodoot Sheet (**Systems Society** award)

- Built a deep learning framework to localize and identify mitotic nuclei from breast histopathological images on **Torch** and extended the idea to leukemia detection.

TEACHING AND VOLUNTEERING EXPERIENCE

Student Lab Representative

Aug 2020 - Oct 2021

Montreal Institute for Learning Algorithms (MILA)

Content Creator, Un/Self-supervised Learning Methods

Aug 2021

Dr. Blake Richards & Dr. Tim Lillicrap, Neuromatch Academy: Deep Learning

Reviewer

IEEE Signal Processing Magazine; IEEE Journal of Biomedical and Health Informatics; Transactions on Medical Imaging; NeurIPS 2022 (Top reviewers); ICLR 2023; NeurIPS 2023;

Teaching Assistant, McGill University

Jan 2019-Apr 2022

Artificial Intelligence; Applied Machine Learning; Computational Perception; Brain-inspired AI

Co-organizer

Computational and Systems Neuroscience (COSYNE) workshop on representation geometry

Mar 2024

Computational Cognitive Neuroscience (CCN) GAC workshop on Neural Representations

Aug 2023

Computational Cognitive Neuroscience (CCN) GAC workshop on Recurrent Networks

Oct 2020

SCHOLARSHIPS AND AWARDS

- **Vanier** Canada Graduate Scholarship, September 2021 (\$50,000/year, 3 years)
- Healthy Brains for Healthy Lives (**HBHL**) Doctoral fellowship, September 2020 (\$15,000)
- 2nd prize in McGill **TechIdea** Pitch Competition, January 2019 (\$250)
- Healthy Brains for Healthy Lives (**HBHL**) Masters fellowship, September 2018 (\$10,000)
- McGill **Faculty of Medicine** Internal fellowship, September 2018 (\$10,000)
- Quebec Bio-Imaging Network (**QBIN**) Foreign Students Scholarship, December 2017 (\$7,000)
- **MITACS** Graduate Fellowship award 2017 for pursuing graduate studies in Canada (\$15,000)
- **MITACS** Globalink Research Internship award 2016 for summer internship 2016 (\$7,500)
- **Honda** Young Engineer and Scientist award, January 2016 (\$3,000)